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To cite this article: Nagham Dawood Salman *et al* 2024 *IOP Conf. Ser.: Earth Environ. Sci.* **1371** 052039

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Evaluation the Antibacterial Activity of some Macro Algae Isolated from Baghdad-Iraq

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Abstract. Because of their aggressiveness and capacity to build biofilms, multidrug-resistant bacteria like *Staphylococcus aureus* present a global threat and cause chronic illnesses. This study focused on two different areas: first, it looked at the antibiotic resistance profiles and biofilm activity of bacteria isolates taken from wounds and newborns. Secondly, it investigated the possibility of using algal extracts as strong agents that combat bacteria and biofilm formation. Two algal extracts, *Spirogyra* and *Oedogonium* (Chlorophyta) methanol extract were examined separately for their antibacterial and antibiofilm properties. The antibacterial and antibiofilm qualities of the algal extract were assessed. Algal extracts may be useful as antibacterial and antibiofilm agents to treat illnesses that are resistant to drugs, according to this study. It is necessary to conduct further research in clinical applications.

Keywords. *Spirogyra*, *oedogonium*, Biofilm, *Staphylococcus aureus*.

1. Introduction

Numerous autotrophic organisms, including algae, may polarize light as a result of photosynthesis, which transforms inorganic substances like carbon dioxide and water into organic matter that stores energy [1]. Because they may create a wide range of secondary metabolites with a wide range of biological functions, macro algae are thought to be a source of bioactive chemicals, brown, red, and green algae have been shown to have compounds having antibacterial, antiviral, antifungal, and antioxidant properties [2].

One of the most common kinds of algae in the world, green algae may be found in a variety of habitats, including freshwater, they also develop quickly in salty water, which makes up around 10% of the total communities, or attached to plants (epiphytic), endolithic rocks, endaphic moist soil, or both [3]. Green algae are known to possess antibacterial qualities that can fight off both gram-positive and gram-negative bacteria [4].

Treating resistant pathogenic bacteria is proving to be an increasingly challenging task [5], since the indiscriminate use of antibiotics often results in infections developing resistance to them, it is important to focus on employing bioactive chemicals to either control or eradicate the germs. An



investigation on the antibacterial qualities of some macroalgae that are found along the Tigris River in Baghdad, Iraq, was conducted in the current study.

2. Material and Methods

2.1. Sample Collection

In the spring of 2022, two genus of macro- algae were isolated, the first alga (*Spirogyra varians*) was collected from Tigris River near the University of Baghdad while the second alga (*Oedogonium sp.*) was isolated from the Al-Baeaytha area along the Tigris River in Baghdad.

In the polythene bags filled with river water, live and healthy macroalgae were collected and transported to the lab.

Running water was used to wash each species to get rid of epiphytes, animal castings, connected detritus, and sand particles. Distilled water was used for the last cleaning, and the species was allowed to dry at room temperature in the shade [4].

2.2. Algal Extract

The active components in macro- algae were extracted using a soxhlet device, according to [4]; within six hours, 20 grams of algal powder and 400 milliliters of Methanol were obtained and extracted. Following the incubator's drying process, the dry extract was weighed and stored in the refrigerator until needed. Preparing various concentrations of algal extracts for every extract, a stock solution was created by, one gram of the extract was dissolved in 2.5 milliliters of the suitable DMSO, and the volume was then increased to five milliliters, to get concentration 200 mg/ml.

2.3. Antibacterial Assay

Algal extracts were subjected to antibacterial testing on Petri plates using the well diffusion technique [6]. The sizes in millimeters of the bacterial growth inhibition halos surrounding the well are used to express the results. DMSO was utilized as the negative control in the absence of macro algal extracts. Clear halos larger than 10 mm were regarded as positive findings; the experimental data in comparison represents the mean $\pm$ SD of each sample. All tests were run in triplicate.

2.4. Assay for the Creation of Biofilm

The biofilm formation test was carried out using the Tissue Culture Plate method, as mentioned by Babour et al. [7].

- Bacterial isolates were grown on B.H.I. agar and incubated for twenty-four hours at 37°C*. Next, 4-5 colonies were suspended in 5 milliliters of regular saline in test tubes and vortexed.
- A 96-well microtiter plate with a flat bottom that was inoculated with 20 μ l of an overnight culture of bacteria with 180 μ l of B.H.I. broth with 2% sucrose. Each isolate was tested in triplicate using 200 μ l of B.H.I. broth containing 2% sucrose in the control wells.
- A lid was placed on the microtiter plate and it was incubated for 24 hours at 37°C.
- After the incubation period, all of the wells' contents were removed, and they were thoroughly cleaned three times using PBS (pH 7.2) before being left to air dry.
- The plates were left to dry for fifteen minutes at room temperature.
- 200 μ l of 1% crystal violet was added to each well, and they were allowed to stand for 15 minutes.
- Following the removal of the crystal violet solution, the wells were allowed to dry at room temperature for three cycles of washing with PBS (pH 7.2) to get rid of the unbounded dye.
- 200 μ l of methanol was used to break the dye's bond with the adhering cells.
- Using an ELISA reader, the absorbance of each well was measured at 630 nm. The control well's O.D. value was subtracted.
- Based on the test's overall O.D. value.
- Each test was run in duplicate three times.

- Four categories were created based on the bacterial test isolates' adhesion ability. Mathur et al. (2006) classified the isolates as follows: the cutoff optical density (OD_c) was defined as being above the mean optical density of the negative control, which consisted only of broth[8].
- The biofilm production was interpreted using the standards provided by [9, 10].
- At 0.1 nm, there is weak or no biofilm formation; at 0.5 nm, there is intermediate to strong biofilm growth. At 0.500 nm, there is a strong to strong biofilm production.

2.5. Evaluation of the *Spirogyra* Extracts' Antibiofilm Activity and Biofilm Inhibitory Percentage

A clear 96-well polystyrene (PS) micro titer plate with a cover that was sterile and flat on the bottom was used to measure the antibiofilm activity of the algal extracts on the five isolated bacteria that were chosen. Pathogens and BHI broth were present in every well [11,12]. After adding and combining the algal extracts, a series of dilutions was used to find the minimum inhibitory concentration (MIC) against the development of biofilms. Mixtures devoid of bacteria were utilized to explain the absorbance of the extract's constituent parts. After the dish was incubated, non-adherent cells were washed out. Sessile bacteria underwent staining, fixing, and washing. At 620 nm, the absorbance was measured. The portion of the biofilm that forms the following formula was used to determine inhibition:

$$\% \text{ Biofilm inhibition} = (\text{OD}_{620} \text{ of the nontreated (control)} - \text{OD}_{620} \text{ of cells treated with algal extracts}) / (\text{OD}_{620} \text{ of the nontreated control}) \times 100$$

The mean standard deviation was calculated when the trials were run in triplicate.

2.6. Mass Spectrometry and Gas Chromatography

Using an Inert cap 1MS high-temperature column (30 m × 0.25 mm id × 0.25 μm film thickness) was acquired for GC-MS analysis from Agilent Technologies (SHIMADZU—Japan). Every sample's derivatization was removed.

The starting column temperature was set at 100°C, while the injector and detector were set at 280°C. The column was filled with a 5 μL sample volume and operated in split (1:10) mode. The oven was heated to 225°C in 1 minute, with hold duration of 4 minutes and a ramp rate of 12.5°C/min. After that, the oven temperature was increased to 300°C with a 5-minute hold period at a ramp rate of 7.5°C/min. Agilent GCMS Solution (SHI-MADZU Japan) and postrun software were used to gather and process the mass spectra. The helium carrier gas was designed to maintain a constant flow rate of 17.5 mL/min. By comparing the compounds' masses with real standards and a search of the NIST library, the compounds were identified [13].

3. Result

3.1. A Structure and Morphologically Characteristics of Algae

In shallow seas, filamentous unbranched green algae belonging to the genus *Spirogyra* sp. create free-floating mats. It is commonly found in sluggish streams, shady littoral zones of lakes, and stagnant waterways like ponds and canals [14]. There are over 400 different species of *Spirogyra* sp. The most distinctive feature of their vegetative structure is their exquisitely coiled chloroplast, which is seen in the cells of algae figure as shown in figure (1) all above agree with [15].



Figure 1. Microscopic examination for *Spirogyra* sp at 40x.

Being photosynthetic, the genus *Spirogyra* makes a significant contribution to the overall amount of carbon dioxide fixed. In their environment, they raise the oxygen content. They are eaten by several aquatic creatures [16]. The algae's slick smoothness is caused by the cell wall, which is made up of an exterior layer of pectin and an interior layer of cellulose Figure (1). The filament *Spirogyra* variance has a length range of 125-200 μm and a breadth range of 28-30 μm . The gametangia's shape is inflated in the center exclusively. Pyrenoids are found in large rows within chloroplasts; they are used to store protein and starch [17].

Oedogonium species, which are filamentous green algae, are categorized as belonging to the Chlorophyta class and consist of 534 species [18]. Within freshwater environments, these species are widely distributed and have a preference for still bodies of water, including small ponds, pools, and ditches along highways, marshes, oxbows, lakes, reservoirs, and rivers [19]. The Oedogoniaceace (Chlorophyta) family of algae includes *Oedogonium Capillare*. Temperature, light intensity, and habitat type all affect the abundance of *Oedogonium* species, all above agree with [20] as shown in figure (2).



Figure 2. Microscopic examination for *oedogonium sp* at 40x.

3.2. Biochemical Analysis Using (GC-MS)

The structure and makeup of the chemicals causing the high antibacterial efficacy of the extract's bioactivity were being investigated by biochemical investigation. As shown in table (3) and figure (3), the identification of natural organic compounds now requires examination using gas chromatography-mass spectrometry (GC-MS). The chemical properties of the *Spirogyra* varians extract were examined using GC-MS techniques in order to determine which compounds were in charge of the biological activity seen.

Table 1. The GC-Mass Spectrophotometry Chromatogram of major compound in the *Spirogyra* varians Methanol Extract.

Number	Rt	Area	Compound
1	7.362	2.88	Benzene, methyl- Toluene
2	53.067	1.62	1- Nonadecane
3	53.919	12.15	Nonadecane
4	54.748	27.47	Hexadecanoic acid, methyl ester
5	60.309	14.73	Heneicosane
6	60.514	5.09	10-Octadecenoic acid, methyl ester
7	60.657	7.45	HEPTADECENE-(8)-CARBONIC ACID
8	61.109	16.95	Methyl stearate
9	62.538	7.20	Eicosane
Total			95.54

Algae are known to contain a wide range of naturally occurring antimicrobial compounds, such as phenols, terpenes, acetogenins, indoles, fatty acids, and volatile halogenated hydrocarbons [21]. Therefore, GC-MS methods were used to examine both fatty acids and volatile compounds in a separated fraction of hexane and ethyl acetate extracts that shown anti-bacterial activity from the researched algae. The result implies that long chain hydrocarbons might serve as a potential bioactive component and be utilized in the creation of pharmaceuticals. Seaweeds are cultivable, which is another advantage for the large-scale production of potential antibacterial chemicals. Further studies

are being carried out to look at the antioxidant and anti-inflammatory qualities of these compounds, as well as the mechanism by which the purified chemicals reduce infections.

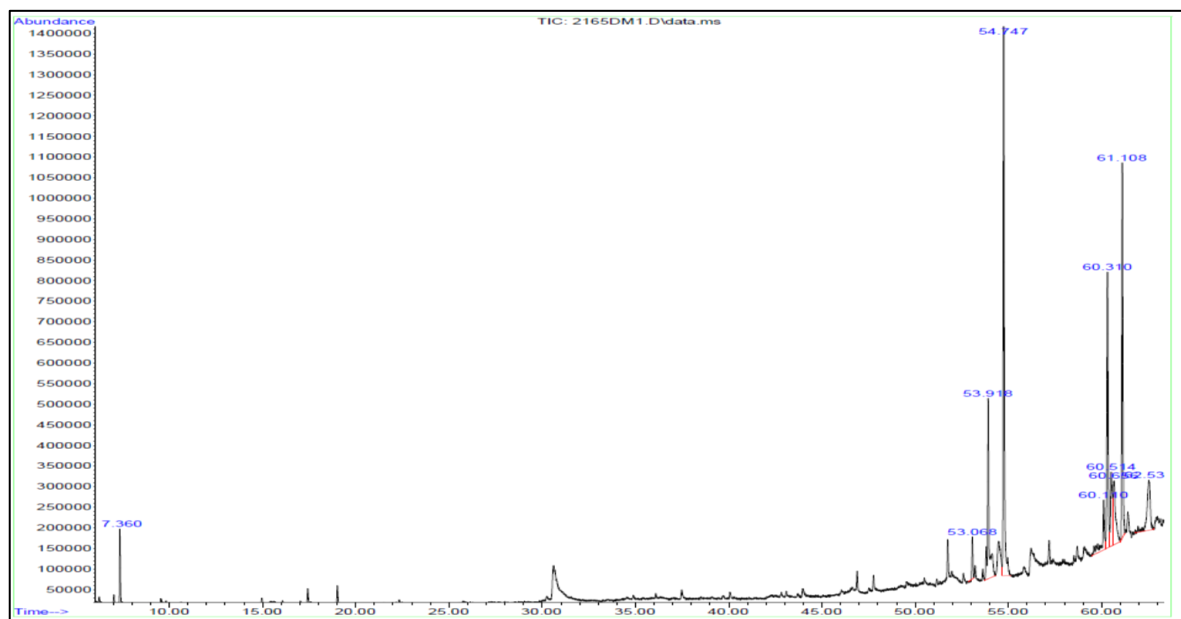


Figure 3. The GC-Mass Spectrophotometry Chromatogram in the *Spirogyra* variants Methanol Extract.

3.3. Antibacterial Activity of Macro-Algae

Using a well diffusion experiment, *Spirogyra varians* from Baghdad University, Iraq were investigated. Table (2) and figure (3) provide a summary of the results of the hot methanol form of the algae extract, which showed varied degrees of antibacterial activity against the test organisms at different doses. It's time to switch to natural antibiotic sources because synthetic antibiotics have potential negative consequences [2]. Producing natural extracts with the same level of efficacy as synthetic antibiotics is imperative. A major risk of biological therapy is the possibility of bacterial resistance to the antibiotic being employed. Based on table (2) *Acinetobacter baumannii* and *Pseudomonas aeruginosa* displayed the greatest inhibitory zone. It was evident that the algal extract in methanol at a concentration of 200 mg/ml had a 34 mm zone of inhibition, which was closer than the other concentrations. Furthermore, *S.epidermidis* had a minimal zone of inhibition, as shown in table (2). It was evident that an extract of algae with an 11 mm zone of inhibition in methanol at a dosage of 50 mg/ml (1). Numerous investigations on the antibacterial activity of algal extracts have been published during the past few decades such as Methanolic extracts of *Chlorella pyrenoidosa*, *Nostoc muscorum*, and *Spirulina platensis* were shown to be effective against human pathogenic bacteria and fungus by algae, which also served as a good source of bacterial resistance[22].

Table 2. *Spirogyra* extract's antimicrobial properties The same-letter means are not statistically different from one another; tiny letters denote vertical comparisons and capital letters denote horizontal comparisons.

Groups		c200	c150	c100	c50	p-value
<i>Acinetobacter baumannii</i>	Mean	34.000 a A	29.666 a B	21.000 b C	19.500 a C	0.001**
	Std. Deviation	3.6055	1.5275	1.7320	.7071	
E.coli	Mean	32.000 a A	23.000 b B	21.000 b B	17.333 b C	0.001**
	Std. Deviation	3.0000	3.0000	2.6457	2.5166	
<i>Pseudomonas aeruginosa</i>	Mean	34.000 a A	30.333 a B	26.333 a C	20.000 a D	0.001**
	Std. Deviation	1.0000	1.5275	1.1547	1.0000	
S.aureus	Mean	26.666 b A	23.666 b A	18.000 c B	15.000 c B	0.001**
	Std. Deviation	1.5275	2.0816	3.0000	2.0000	
<i>S.epidermidis</i>	Mean	24.666 b A	20.000 b B	15.333 c C	11.666 c D	0.001**

Groups		c200	c150	c100	c50	p-value
	Std. Deviation	2.5166	1.0000	2.5166	1.5275	
	p-value	0.002**	0.001**	0.002**	0.002**	-

Despite the results obtained from the *Spirogyra* extract, this contradicts the effect of the *Oedogonium*, which did not show any effect on all types of bacteria used in this research. This is due to several factors, As seen in figure (3B), a number of factors can impact the antimicrobial activity of algal extracts, including the type of algae [22], the extraction method, the testing protocol, the solvent used in the extraction [23], the season or time of year the samples were collected, and the location of the sample collection [24].

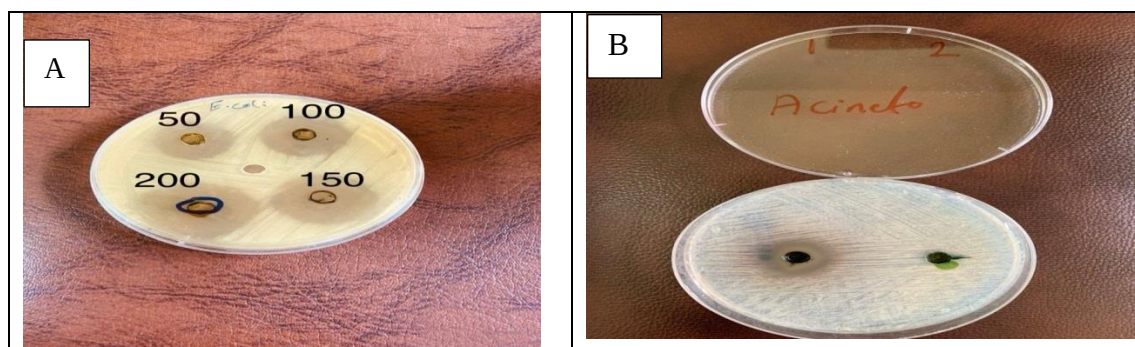


Figure 4. (A) antibacterial activity of crude extract *Spirogyra varians* against *E.coli* at different concentrations ;(B) The difference between *Spirogyra* effect and *Oedogonium* on *Acinetobacter baumannii*, Where the number 1 represents *Spirogyra* extract effect While the number 2 represents *Oedogonium* extract effect.

3.4. Biofilm Formation Assay

According to Hassan et al. 2011 and Stepanovi et al. 2007 method [9, 10], five types of bacteria were tested for their ability to produce biofilm, and the result was shown that only *S.aureus* was able to formation strong biofilm under the influence of these conditions as shown in the table (3) while *S.epidermidis* and *Pseudomonas aeruginosa* have a less ability to form biofilm moderate.

Table 3. Biofilm Producing Ability (BPA) according to Hassan et al. 2011 and Stepanovi et al. 2007
OD Less than 0.2 weak, from 0.2- 0.5 moderate , more than 0.5 strong.

Bacteria	Biofilm Producing Ability (BPA)	
<i>E.coli</i>	OD 0.151	Weak
<i>Acinetobacter baumannii</i>	OD 0.193	Weak
<i>Pseudomonas aeruginosa</i>	OD 0.215	Moderate
<i>S.aureus</i>	OD 0.603	Strong
<i>S.epidermidis</i>	OD 0.308	Moderate

Despite the conditions that were provided, the ability of *Acinetobacter baumannii* and *E.coli*. bacteria to form biofilm was weak or almost non-existent All types of bacteria were treated with three different concentrations of the extract, 200mg /ml, 100mg /ml, and 50mg/ml , it was found that there is a high rate of inhibition of the biofilm formation . It was found that concentration 50 is the least effective concentration in inhibition, as there is a clear difference between it and the rest of the concentrations, and this confirms that When the concentration of the extract increases, its effectiveness increases more than from lower concentration, because the active ingredients present begin to diminish.

Table 4. Effect of *Spirogyra* extracts at three different concentration on the inhibition of biofilm formation.

	O.D Before Treatment (control)	O.D after treatment at 200mg/ml	Inhibition rate %= (od control- o.d after/o.d control)×100	O.D after treatment at 100mg/ml	Inhibition rate %	50mg/ml O.D after treatment at	Inhibition rate %
<i>Acintobacter</i>	0.193	0.048	75	0.071	62	0.145	30
<i>S.aureus</i>	0.603	0.056	90	0.154	74	0.439	27
<i>Pseudumona</i>	0.215	0.032	85	0.055	74	0.11	48
<i>S.epidermus</i>	0.308	0.054	82	0.098	68	0.221	29
<i>E.coli</i>	0.151	0.053	65	0.078	48	0.127	16

A series of dilutions were made of the extract on the formation of the biofilm. It was found, according to the table (4) that as the concentration of the extract increased, the formation of the biofilm decreased or was absent.

4. Discussion

The most promising method of treating these sorts of diseases appears to be the use of natural chemicals like medicinal alga, while antibiotic resistance among dangerous bacteria is spreading. Numerous studies on this topic and the effects of medicinal algae on various bacteria, such as *S. aureus*, *E. coli*, *Pseudomonas aeruginosa*, and others, have been done to date. This study has demonstrated that medicinal algae, including *Cladophora glomerata*, *Chara*, *Spirulin platensis*, and many others, can alter the structure of bacterial cells wall and inhibiting their growth.

In 2016, Dawish et al. studies indicated the antimicrobial effects of *Spirogyra variant* on multi drug resistant bacterial pathogens by using chloroform to extract the active substances from the *Spirogyra*, while in our study the extraction process was carried out using methanol, and we obtained similar results [14].

Also in 2022, Al-Khazraji, R.A.A. conducted research using methanolic extract [22], and this is similar to what she reported in our study, but there are some differences. It was found that the diameter of inhibition in our study for some types of bacteria is higher than the diameter of inhibition in the first study, and this is due to several factors, the most important of which is the season of sample collection, the location of collection, and many other reasons.

As for the process of inhibiting the biofilm formation of *Staph* bacteria, there are many similar studies using plants or algae, and this is what was reported by [26], used two types of plants that demonstrated their ability to inhibit biofilm formation.

The effectiveness of *Oedogonium capillare* extracts as an antibacterial agent against 23 distinct bacterial species belonging to the Enterobacteriaceae, Pseudomonadaceae, Aeromonadaceae, and Vibrionaceae families was demonstrated in vitro by [27]. However, this result literally contradicts what was obtained in our study, as we did not find that this alga had any effect on all types of bacteria isolated in this research, and this is due to many reasons, but apparently the most important reason is the time of collecting the samples.

Conclusions

Promising antibacterial and antibiofilm properties were demonstrated by two distinct algal extracts: the methanol extracts from *Oedogonium Capillare* and *Spirogyra* variants. When compared to *Oedogonium* extracts, the *Spirogyra* extract exhibited better effects, which led to a notable rate of inhibition on *S. aureus* biofilm development. The fatty acid content of the algal extracts as well as particular substances like heptadecene -(8)-carbonic acid and methyl stearate were thought to be responsible for the effects that were seen. The physicochemical characteristics of the algal extracts were found to be advantageous for their possible application as antibacterial agents, as per the GC-MS analysis. These phytochemical compounds' potential for use in clinical settings is further supported by the fact that they exhibit intriguing drug-like and pharmacokinetic features. It is promising that the algal extracts may effectively combat resistant strains of bacteria and biofilms, implying the possibility

of creating alternative therapeutic approaches. The combined algae extracts' synergistic activity emphasizes their potential for treating drug-resistant illnesses and begs for more research.

Acknowledgements

This study was conducted with the supports of Biological Sciences College of Science, University of Mustansiriyah. Baghdad-Iraq.

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